M. Thufail Alwannabil Samas

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SUMMARY

Mathematics graduate and certified Data Scientist with practical experience in data analytics, machine learning, Python-based data processing, and dashboard development. Built analytical projects across forecasting, classification, and monitoring use cases, supported by published research on XGBoost-based rainfall prediction.

EDUCATION

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA) — **Surabaya, Indonesia**
Bachelor of Mathematics (B.Math.) — August 2021 – January 2025

- **GPA:** 3.46/4.00
- **Undergraduate Thesis:** Rainfall forecasting in Sumenep Regency using Extreme Gradient Boosting (XGBoost) with Grid Search hyperparameter tuning.
- **Academic Highlight:** Completed bachelor's degree in 7 semesters (3.5 years)
- **Relevant Coursework:** Artificial Intelligence, Programming, Statistical Methods, Mathematical Statistics, Multivariate Analysis, Sampling Techniques, Numerical Methods, Applied Linear Algebra.

PROFESSIONAL EXPERIENCE

PT Aplikanusa Lintasarta — **Surabaya, Indonesia**
Junior Programmer — January 2026 – April 2026

- Developed a project-stage DWDM optical monitoring and assurance analytics system to process PRTG telemetry from approximately 400 optical sensors, combining 5-minute signal data, daily summaries, downtime, coverage, and sensor/link metadata.
- Built a Python and PostgreSQL time-series data pipeline for metadata synchronization, telemetry ingestion, preprocessing, gap handling, processed data generation, event detection, and API-served outputs across varying historical telemetry windows.
- Implemented data-quality and failure-detection logic to identify missing telemetry, zero coverage, downtime indicators, empty rows, low optical power values, bounded single-bucket gaps, and contiguous failure events from processed time-series data.
- Developed robust statistical time-series detectors for change-point and gradual degradation analysis using median baselines, MAD-based noise estimation, Theil-Sen slope, Mann-Kendall trend testing, persistence checks, stability scoring, and step-dominance filtering.
- Built a Python-backed HTML/CSS/JavaScript dashboard to visualize active failures, change-point and degradation events, investigation charts, and P1–P10 sensor priorities based on latest optical signal degradation relative to historical best levels.

SMP GIKI 3 Surabaya — **Surabaya, Indonesia**
Coding and AI Teacher — August 2025 – December 2025

- Taught Python fundamentals to Grade 8 and 9 students, covering variables, data types, input/output, operators, conditionals, loops, and basic programming logic.
- Introduced foundational AI concepts using browser-based tools such as Teachable Machine, helping students understand datasets, model training, and image classification.
- Designed lesson materials, coding exercises, quizzes, and sample code from scratch, and evaluated students through practical programming activities and quizzes.

Stasiun Meteorologi Maritim Kelas II Tanjung Perak Surabaya (BMKG) — **Surabaya, Indonesia**
Data Science Intern — February 2024 – June 2024

- Developed a one-page extreme weather early warning dashboard using HTML, CSS, JavaScript, PHP, MySQL, and API-based periodic updates.
- Designed the dashboard to visualize current weather conditions, warning indicators, tables, charts, filters, and monitoring views for BMKG’s internal forecasting workflow.
- Processed and compared BMKG cloud prediction imagery against actual Himawari IR satellite imagery using Python, supporting descriptive validation of cloud movement and temperature patterns.

- Developed and documented a BiLSTM-based rainfall forecasting model for a final internship research paper, with the best configuration achieving a MAAPE of 0.8073 and an RMSE of 10.2734.

Universitas Islam Negeri Sunan Ampel Surabaya (UINSA) **Surabaya, Indonesia**
Programming Teaching Assistant August 2022 – July 2024
- Assisted in delivering Basic Programming and Programming courses across four semesters, supporting undergraduate students in MATLAB, Python, and computational problem-solving.
- Guided students in programming fundamentals, debugging, data handling, and algorithmic thinking to help translate mathematical and computational concepts into working code.
- Developed introductory machine learning materials on linear regression, K-Nearest Neighbors, and K-Means clustering, and provided structured feedback on student assignments.

PROJECTS

Rainfall Forecasting with XGBoost Regressor
Built a daily rainfall forecasting workflow using BMKG weather features, historical rainfall, 1-day lag, 10-fold TimeSeriesSplit, Min-Max normalization, and Grid Search, selecting the best configuration with MAAPE of 0.9152 and RMSE of 11.9566.

Rainfall Forecasting with BiLSTM
Built a daily rainfall forecasting workflow using BMKG weather features, historical rainfall, 7-day lag, 80:20 chronological split, Min-Max normalization, and Grid Search, selecting the best configuration with MAAPE of 0.8073 and RMSE of 10.2734.

Rice Price Forecasting with ARIMA
Built a monthly rice price forecasting workflow for medium and premium rice using ARIMA, 80% initial training history, walk-forward validation, ADF differencing, ACF/PACF inspection, residual diagnostics, and comparison with reference models, achieving MAPE of 2.081% and 1.8174%.

Thyroid Cancer Recurrence Classification with XGBoost Classifier
Built a thyroid cancer recurrence classification workflow using XGBoost, Information Gain, 10-fold K-Fold Cross Validation, Grid Search, and default-model comparison, selecting the best configuration with an accuracy of 0.9725, 0.9537 F1-score, 0.9805 specificity, and 0.9860 ROC-AUC.

PUBLICATIONS

Prediksi Curah Hujan di Kabupaten Sumenep Menggunakan Metode Extreme Gradient Boosting (XGBoost) dan Algoritma Grid Search
M. Thufail Alwannabil Samas, Nurissaidah Ulinnuha, Moh. Hafiyusholeh
MIND (Multimedia Artificial Intelligent Networking Database) Journal, Vol. 11, No. 1, pp. 30–43, June 2026. DOI: 10.26760/mindjournal.v11i1.30-43

CERTIFICATIONS

Badan Nasional Sertifikasi Profesi (BNSP) **Indonesia**
Data Scientist September 2024 – September 2027
Competency coverage: business objective definition, data science technical goal setting, data analysis, data validation, data cleaning, data construction, model scenario development, model building, model evaluation, and modeling review.

TECHNICAL SKILLS

- **Programming:** Python, R, MATLAB, PHP | **Query & Databases:** SQL, PostgreSQL, MySQL
- **Web Tech:** REST API, HTML, CSS, JavaScript | **Development Tools:** Jupyter Notebook, Git, GitHub
- **Libraries:** NumPy, Pandas, Scikit-learn, XGBoost, Statsmodels, TensorFlow/Keras, Matplotlib, Seaborn
- **Data Analytics & Modeling:** Data Analysis, Data Preprocessing, Feature Engineering, Statistical Modeling, Machine Learning, Deep Learning